SAT Math Mastery · Session 5

Equivalent Expressions and Exponential Functions

Time 3 hours: Part A in class (about 2 hours of solving), then Part B, the self-assessment, done independently (about 50 minutes), and a short error log.

Topics equivalent expressions: exponent rules, rational expressions, factoring and identities · radical equations · systems with exponents and radicals · radical functions and domain · exponential growth and decay models · interpreting exponential models · recursive and geometric sequences · successive percent changes

Name / Date _____ _____

PART A · IN CLASS — 35 PROBLEMS

Each problem states its tool. Write your work in the space below. Multiple choice: one correct option. Student-produced response (SPR): enter an exact value (fraction or decimal). Grey boxes show exactly what to type in Desmos, one expression per line; tap a grey dot to read an intersection, intercept or vertex. A letter such as a, c or k becomes a slider when you accept Desmos's offer to add one.

1. HARD · DESMOS

For all real values of x,

$$\frac{6^{2x+3}}{2^{2x-1}} = k \cdot 9^x$$

where k is a constant. What is the value of k?

1. Move k to one side and type `g(x)=6^(2x+3)/(2^(2x-1)*9^x)`. The x terms cancel, so Desmos draws a flat line and k is its height.
2. Tap the line where it crosses the y-axis: (0, 432), so $k = 432$.
3. Check a second point on the same line. Type `g(2)` : it returns 432: the same height.

Answer: 432

2. HARD · DESMOS

$$(x+p)(x+q) = x^2 + bx + c$$

In the equation above, p and q are real constants and b and c are integers. Which of the following must be true?
I. $p+q$ is an integer
II. p^2+q^2 is an integer
III. $\frac{p^2+q^2}{2}$ is an integer

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Test one case with integers $b = 3$, $c = 2$: type `p=(3+sqrt(3^2-4*2))/2` (so $p = 2$) and `q=(3-sqrt(3^2-4*2))/2` (so $q = 1$).
2. Type `p+q` : it returns 3 and type `p^2+q^2` : it returns 5: both integers, matching statements I and II (algebra shows they always are, since $b = p+q$ and $p^2+q^2 = b^2 - 2c$).
3. Type `(p^2+q^2)/2` : it returns 2.5: not an integer, so statement III does not have to be true.
4. Only I and II: option (B).

Answer: (B)

3. HARD · DESMOS

For all $x > 1$,

$$\frac{a}{x-1} - \frac{bx+c}{x^2+x+1} = \frac{3x^2+8x+4}{x^3-1}$$

where a, b, and c are constants. What is the value of c?

1. Multiply by x^3-1, then put $x=1$ so that the c and b terms vanish: Type `(3*1^2+8*1+4)/(1^2+1+1)` : it returns 5, so $a=5$.
2. Put $a=5$ back and isolate $bx+c$: type `g(x)=(3x^2+8x+4-5(x^2+x+1))/(-(x-1))` . Desmos draws a straight line (with a hole at $x=1$).
3. Tap the y-intercept: Type `g(0)` : it returns -1, so $c=-1$.
4. The line also passes through $(2,3)$: Type `g(2)` : it returns 3, so the slope is $b=2$.

Answer: -1

4. HARD · DESMOS

For $y \neq 0$ and $y \neq \frac{1}{3}$, which of the following is equivalent to

$$\frac{\frac{1}{y^3} - 27}{\frac{1}{y} - 3} \ ?$$

(A) $\frac{1}{y^2} - \frac{3}{y} + 9$

(B) $\frac{1}{y^2} + 9$

(C) $\frac{1}{y^2} + \frac{6}{y} + 9$

(D) $\frac{1}{y^2} + \frac{3}{y} + 9$ ✓

1. Type the original expression: `f(x)=(1/x^3-27)/(1/x-3)` . At $y=0.5$: Type `f(0.5)` : it returns 19.
2. Now try each option at $y=0.5$: Type `1/0.5^2-3/0.5+9` : it returns 7; type `1/0.5^2+9` : it returns 13; type `1/0.5^2+6/0.5+9` : it returns 25; type `1/0.5^2+3/0.5+9` : it returns 19.
3. Only `y=1/x^2+3/x+9` matches 19; graphing it on top of f gives the same curve. Option (D).

Answer: (D)

5. HARD · DESMOS

The polynomial $p(x) = x^3 - 9x^2 + kx - 12$ has a positive integer constant k. For some positive integer n, $x-n$ is a factor of $p(x)$. What is the sum of all possible values of k?

1. A factor $x-n$ means $p(n)=0$, so $k = \frac{12}{n} + 9n - n^2$, and n must divide 12. Type `k(n)=12/n+9n-n^2` .
2. Try the divisors: Type `k(1)` : it returns 20; type `k(2)` : it returns 20; type `k(3)` : it returns 22; type `k(4)` : it returns 23.
3. Continue: Type `k(6)` : it returns 20; type `k(12)` : it returns -35. Reject $n=12$ (negative).
4. The distinct positive values are 20, 22, 23 (count 20 once): Type `20+22+23` : it returns 65.

Answer: 65

6. HARD · DESMOS

$$P = \frac{a}{\sqrt[5]{w^2}}$$

The formula above relates the positive quantities P and w, where a is a positive constant. Which of the following expresses w in terms of P and a?

(A) $w = \left(\frac{a}{P}\right)^{5/2}$ ✓

(B) $w = \left(\frac{P}{a}\right)^{5/2}$

(C) $w = \left(\frac{a}{P}\right)^{2/5}$

(D) $w = \left(\frac{P}{a}\right)^{2/5}$

1. Test with $a = 2$ and $w = 5$: `P=2/(5^2)^(1/5)` (so $P = 1.0506$).

2. Evaluate two options: Type `(2/P)^(5/2)` : it returns 5; type `(P/2)^(2/5)` : it returns 0.773.

3. And the other two: Type `(P/2)^(5/2)` : it returns 0.2; type `(2/P)^(2/5)` : it returns 1.2937.

4. Only $\left(\frac{a}{P}\right)^{5/2}$ returns $w = 5$: option (A).

Answer: (A)

7. HARD · DESMOS

For all real numbers x,

$$8^x \cdot 4^{x/3} \cdot \left(\sqrt{2}\right)^x = b^x,$$

where b is a positive constant. Which of the following is equal to 2?

(A) $b^{6/25}$ ✓

(B) $b^{25/6}$

(C) b^{25}

(D) $b^{-6/25}$

1. Add the exponents of 2: Type `3+2/3+1/2` : it returns 4.1667, which is $\frac{25}{6}$. So $b = 2^{25/6}$.

2. Check: Type `8*4^(1/3)*sqrt(2)` : it returns 17.9594; type `2^(25/6)` : it returns 17.9594 (the same number, b).

3. `b=2^(25/6)` (so $b = 17.9594$). Type `b^(6/25)` : it returns 2; type `b^(-6/25)` : it returns 0.5.

4. Only $b^{6/25}$ equals 2: option (A).

Answer: (A)

8. HARD · DESMOS

$$\frac{2}{a}-\frac{3}{b}=\frac{5}{c}$$

In the equation above, a, b and c are nonzero constants and $2c \neq 5a$. Which of the following expresses b in terms of a and c?

(A) $b=\dfrac{3ac}{5a-2c}$

(B) $b=\dfrac{2c-5a}{3ac}$

(C) $b=\dfrac{3ac}{2c+5a}$

(D) $b=\dfrac{3ac}{2c-5a}$ ✓

1. Choose $a=1$ and $b=2$ and find the c that fits: Type `5/(2/1-3/2)` : it returns 10, so $c=10$.
2. Now test the options with $a=1$, $c=10$ (they should return $b=2$): Type `3*1*10/(2*10-5*1)` : it returns 2; type `3*1*10/(5*1-2*10)` : it returns -2.
3. Type `(2*10-5*1)/(3*1*10)` : it returns 0.5; type `3*1*10/(2*10+5*1)` : it returns 1.2.
4. Only $\dfrac{3ac}{2c-5a}$ returns 2: option (D).

Answer: (D)

9. HARD · DESMOS

The polynomial $6x^4+kx^2+24$, where k is a constant, can be written as $(2x^2+a)(3x^2+b)$, where a and b are integers. What is the greatest possible value of k?

1. Expanding gives $ab=24$ and $k=2b+3a$. Type `k(a)=2(24/a)+3a` .
2. Try the divisors of 24: Type `k(1)` : it returns 51; type `k(2)` : it returns 30; type `k(3)` : it returns 25; type `k(4)` : it returns 24.
3. Continue: Type `k(6)` : it returns 26; type `k(8)` : it returns 30; type `k(12)` : it returns 40; type `k(24)` : it returns 74. Negative pairs give negative k.
4. The greatest value is the last one: $k=74$.

Answer: 74

10. HARD · DESMOS

For a real number a with $|a| \neq 1$,

$$\frac{1}{1-a}+\frac{1}{1+a}+\frac{2}{1+a^2}+\frac{4}{1+a^4}=10.$$

What is the value of a^8?

(A) $\dfrac{5}{4}$

(B) $\dfrac{9}{5}$

(C) $\dfrac{1}{5}$ ✓

(D) $\dfrac{4}{5}$

1. Type the left side `y=1/(1-x)+1/(1+x)+2/(1+x^2)+4/(1+x^4)` and the line `y=10` .
2. Tap the intersection (zoom in near $x=0.8$): (0.8178, 10), so $a \approx 0.81777$.
3. Type `0.81777^8` : it returns 0.2, so $a^8=\dfrac{1}{5}$: option (C).

Answer: (C)

11. HARD · NO CALCULATOR NEEDED

For real numbers a and b, $a^2 - b^2 = 36$. What is the value of $(3a + b)^2 - (a + 3b)^2$?

1. Pick numbers with $a^2 - b^2 = 36$, e.g. $a = 10$, $b = 8$: Type `(3*10+8)^2-(10+3*8)^2` : it returns 288.

2. Another pair, $a = 6$, $b = 0$: Type `(3*6+0)^2-(6+3*0)^2` : it returns 288 (same value).

3. Algebra explains it: the expression is $8(a^2 - b^2)$. Type `8*36` : it returns 288.

Answer: 288

12. HARD · DESMOS

If $2x + 3y = 8$, which of the following is equivalent to $2x^2 + 3xy - 4x - 6y$?

(A) $8x - 8$ (B) $16 - 8x$

(C) $8x - 4y$ **(D)** $8x - 16$ ✓

1. If $2x + 3y = 8$, then $(x, y) = (1, 2)$ works. The original expression there: Type `2*1^2+3*1*2-4*1-6*2` : it returns -8.

2. Test the options at $(1, 2)$: Type `8*1-4*2` : it returns 0; type `8*1-16` : it returns -8; type `8*1-8` : it returns 0; type `16-8*1` : it returns 8.

3. Only $8x - 16$ matches. Check with $(x, y) = (4, 0)$: Type `2*4^2+3*4*0-4*4-6*0` : it returns 16; type `8*4-16` : it returns 16. Option (D).

Answer: (D)

13. HARD · DESMOS

Let t be a positive real number with $t \neq 1$. Define $m = \sqrt[3]{t^2}$ and $n = \sqrt{t}$. If

$$\frac{m^p \cdot n^3}{\sqrt[6]{t}} = t^5,$$

what is the value of p?

1. Test with $t = 2$: type `y=(2^(2/3))^x*(sqrt(2))^3/2^(1/6)` and `y=2^5` (letting x stand for p).

2. Tap the intersection: (5.5, 32), so $p = 5.5$.

3. Check: Type `2^(2*5.5/3+3/2-1/6)` : it returns 32, which is 2^5.

4. As a fraction, $5.5 = \frac{11}{2}$.

Answer: $\frac{11}{2}$ **or** 5.5

14. HARD · DESMOS

The expression $\dfrac{(2x-3)^4(x+2)-7}{5}$ is equivalent to $ax^5+bx^4+cx^3+dx^2+ex+f$, where a, b, c, d, e and f are constants. What is the value of $a+b+c+d+e+f$?

(A) $-\dfrac{4}{5}$ ✓ (B) $\dfrac{3}{5}$

(C) -4 (D) 31

1. The sum of the coefficients is the value at $x=1$. Type `f(x)=((2x-3)^4(x+2)-7)/5`.
2. Type `f(1)`: it returns -0.8, which is $-\dfrac{4}{5}$.
3. Compare with the options: Type `-4/5`: it returns -0.8; type `3/5`: it returns 0.6; type `-4`: it returns -4. Only the first matches: option (A).

Answer: (A)

15. HARD · DESMOS

The polynomial $12x^4-84x^2+144$ can be written in the form $(ax^2+b)(cx^2+d)$, where a, b, c and d are integers with $a>0$ and $c>0$. What is the least possible value of $a+b+c+d$?

(A) -27 (B) -22

(C) -18 **(D) -38 ✓**

1. Check the factoring at $x=1$: Type `12*1^4-84*1^2+144`: it returns 72; type `(1^2-3)*(12*1^2-48)`: it returns 72. So $(x^2-3)(12x^2-48)$ works, i.e. $12(x^2-3)(x^2-4)$ split as $(ax^2-3a)(cx^2-4c)$, $ac=12$.
2. The sum is $-2a-3c$ with $c=\dfrac{12}{a}$. Type `s(a)=-2a-3(12/a)` and evaluate: Type `s(1)`: it returns -38; type `s(2)`: it returns -22; type `s(3)`: it returns -18; type `s(4)`: it returns -17.
3. Type `s(6)`: it returns -18; type `s(12)`: it returns -27. Swapping the factors, type `t(a)=-3a-2(12/a)` and type `t(12)`: it returns -38.
4. The least value is -38: option (D).

Answer: (D)

16. HARD · DESMOS

The function f is defined by $f(x)=\dfrac{6^{x+1}\cdot 4^x}{3^{x-2}}$. It can be written in the form $f(x)=a\cdot b^{x-1}$, where a and b are positive constants. What is the value of $a+b$?

(A) 62 **(B) 440 ✓**

(C) 456 (D) 3456

1. In the form $a\cdot b^{x-1}$, $a=f(1)$. Type `f(x)=6^(x+1)*4^x/3^(x-2)` and type `f(1)`: it returns 432.
2. The ratio of consecutive values is b: Type `f(2)/f(1)`: it returns 8.
3. So $a+b=$ type `432+8`: it returns 440: option (B).

Answer: (B)

17. HARD · DESMOS

For a positive real number m, let $a = \sqrt[3]{m}$ and $b = \sqrt[4]{m}$. If $ab = 128$, what is the value of ab^2?

1. Type `y=x^(1/3)*x^(1/4)` and `y=128` (letting x stand for m).

2. Tap the intersection: (4,096, 128), so $m = 4096$.

3. Now ab^2 with $a = m^{1/3}$, $b = m^{1/4}$: Type `4096^(1/3)*(4096^(1/4))^2`: it returns 1,024.

Answer: 1,024

18. HARD · DESMOS

For all real numbers x, the function g is defined by $g(x) = \dfrac{9^{x+2} \cdot 10^{x+1}}{6^{x-1}}$. The function can be written in the form $g(x) = k \cdot 15^{x+1}$, where k is a constant. What is the value of k?

1. Divide g by 15^{x+1} and graph: `g(x)=9^(x+2)*10^(x+1)/6^(x-1)` and `h(x)=g(x)/15^(x+1)`. Desmos draws a flat line, so k is its height.

2. Tap the line at the y-axis: Type `g(0)/15^(0+1)`: it returns 324, so $k = 324$.

3. Check at $x = 3$: Type `g(3)/15^(3+1)`: it returns 324.

Answer: 324

19. HARD · DESMOS

Positive real numbers a and b, both greater than 1, satisfy $a^3 = b^2$. If $\dfrac{a^9}{\sqrt[3]{b}} = b^k$, what is the value of k?

(A) $\dfrac{17}{3}$ ✓ (B) 6

(C) $\dfrac{19}{3}$ (D) $\dfrac{17}{2}$

1. Pick $b = 8$. Then $a^3 = 64$, so Type `8^(2/3)`: it returns 4 gives $a = 4$.

2. Evaluate the left side: Type `4^9/8^(1/3)`: it returns 131,072.

3. Solve $8^k = 131072$: Type `ln(131072)/ln(8)`: it returns 5.6667, which is $\dfrac{17}{3}$: type `17/3`: it returns 5.6667. Option (A).

Answer: (A)

20. HARD · DESMOS

Positive real numbers x and y satisfy $x \neq y$, $x + y = 15$ and $xy = 36$. What is the value of

$$\frac{\frac{1}{x} - \frac{1}{y}}{\frac{1}{x^2} - \frac{1}{y^2}} \; ?$$

(A) $\frac{5}{12}$

(B) 1

(C) 4

(D) $\frac{12}{5}$ ✓

1. Type `y=x^2-15x+36` (from $t^2 - 15t + 36$).

2. Tap the zeros: (3, 0) and (12, 0), so $\{x, y\} = \{3, 12\}$.

3. Evaluate: Type `(1/12-1/3)/(1/12^2-1/3^2)` : it returns 2.4, which is type `12/5` : it returns 2.4. Option (D).

Answer: (D)

21. HARD · DESMOS

What is the solution set of the equation

$$\sqrt{x + 7} = x - 5?$$

(A) $x = 2$ only

(B) $x = 9$ only ✓

(C) $x = 2$ and $x = 9$

(D) There is no solution.

1. Graph each side: type `y=sqrt(x+7)` and `y=x-5` .

2. Tap the intersection: (9, 4). The line and curve meet only once, so $x = 9$.

3. Type `sqrt(9+7)` : it returns 4; type `9-5` : it returns 4. Both sides agree at $x = 9$.

4. Squaring also produces $x = 2$, but there the line is at -3 and the curve at 3: extraneous. Option (B).

Answer: (B)

22. HARD · DESMOS

a and b are positive consecutive integers, and $b > a$. If b is thirteen more than $\sqrt{a}$, what is the value of a?

1. Consecutive integers: $b = a + 1$, so $a + 1 = 13 + \sqrt{a}$. Type `y=x+1` and `y=13+sqrt(x)` (letting x stand for a).

2. Tap the intersection: (16, 17), so $a = 16$ (the curves cross once).

3. Check: Type `13+sqrt(16)` : it returns 17, which equals $16 + 1$.

Answer: 16

23. HARD · DESMOS

Let m be an even integer. How many possible values of m satisfy $\sqrt{m+11} \le 4$?

1. Type `y=sqrt(x+11)` and `y=4` (letting x stand for m).

2. The curve starts at $(-11, 0)$ and meets the line at $(5, 4)$. It is at or below the line for $-11 \le m \le 5$.

3. The even integers there run from -10 to 4: Type `(4-(-10))/2+1`: it returns 8 values.

Answer: 8

24. HARD · DESMOS

Real numbers x and y satisfy

$$\sqrt[3]{2^x} \cdot 4^y = 256 \qquad \text{and} \qquad \frac{16^x}{\sqrt[3]{4^y}} = 2^{22}.$$

What is the value of xy?

1. Write both equations with base 2 and type the exponent equations: `x/3+2y=8` and `4x-2y/3=22` (letting x, y be the unknowns).

2. Tap the intersection: $(6, 3)$, so $x = 6$, $y = 3$.

3. Type `6*3`: it returns 18.

4. Check the originals: Type `(2^6)^(1/3)*4^3`: it returns 256; type `16^6/(4^3)^(1/3)`: it returns 4,194,304; type `2^22`: it returns 4,194,304.

Answer: 18

25. HARD · DESMOS

Positive real numbers x and y satisfy

$$\sqrt{x} - \sqrt{y} = 2 \qquad \text{and} \qquad x + y = 26.$$

What is the value of xy?

1. Type `sqrt(x)-sqrt(y)=2` and `x+y=26` (curves in the xy-plane).

2. Tap the intersection: (19.9282, 6.0718), i.e. $x \approx 19.9282$, $y \approx 6.0718$.

3. Type `19.9282*6.0718`: it returns 121, so $xy = 121$.

Answer: 121

26. HARD · DESMOS

The function $m(z) = -\sqrt{z^2 + pz + q}$ has graph passing through $(6, 0)$ and $\left(0, -\sqrt{252}\right)$, where p and q are constants. What is the greatest value of t for which $m(t) = 0$?

1. The point $(0, -\sqrt{252})$ gives $q = 252$. The point $(6, 0)$ gives $36 + 6p + 252 = 0$: Type `-(36+252)/6`: it returns -48, so $p = -48$.
2. Zeros of m are zeros of $z^2 - 48z + 252$: type `y=x^2-48x+252`.
3. Tap the zeros: (6, 0) and (42, 0). The greatest is 42.
4. Check the product: Type `6*42`: it returns 252.

Answer: 42

27. HARD · DESMOS

The function $h(x) = -\sqrt{x^2 + bx + c}$ has a graph passing through $(2, 0)$ and $\left(0, -\sqrt{314}\right)$, where b and c are constants. What is the distance between the two x-intercepts of the graph?

1. The point $(0, -\sqrt{314})$ gives $c = 314$. The point $(2, 0)$ gives $4 + 2b + 314 = 0$: Type `-(4+314)/2`: it returns -159, so $b = -159$.
2. Type `y=x^2-159x+314`.
3. Tap the two zeros: (2, 0) and (157, 0).
4. The distance between the intercepts: Type `157-2`: it returns 155.

Answer: 155

28. HARD · DESMOS

The function $f(x) = 2\left(3x + 1 - \sqrt{5}\right)\left(3x + 1 + \sqrt{5}\right)$ can be written as $f(x) = ax^2 + bx + c$, where a, b and c are constants. What is the value of $\dfrac{b + c}{a}$?

1. Type `f(x)=2(3x+1-sqrt(5))(3x+1+sqrt(5))`. Then $f(0) = c$: Type `f(0)`: it returns -8.
2. Evaluate $f(1) = a + b + c$ and $f(-1) = a - b + c$: Type `f(1)`: it returns 22; type `f(-1)`: it returns -2.
3. Then $a = \dfrac{f(1) + f(-1)}{2} - c$ and $b = \dfrac{f(1) - f(-1)}{2}$: Type `(22+(-2))/2-(-8)`: it returns 18; type `(22-(-2))/2`: it returns 12.
4. Finally Type `(12+(-8))/18`: it returns 0.2222, which is $\dfrac{2}{9}$.

Answer: $\dfrac{2}{9}$ **or** 0.2222

29. HARD · DESMOS

The polynomial function p is defined by $p(x) = -(x+3)^2(x-1)^3(x-4)^2$. Consider the following statements.
I. $p(x) \geq 0$ for all $x < 1$.
II. The graph of $y = p(x)$ crosses the x-axis at $x = 4$.
III. $p(x) \to \infty$ as $x \to -\infty$.
Which of the statements must be true?

(A) I only

(B) I and II only

(C) I and III only ✓

(D) II and III only

1. Type `p(x)=-(x+3)^2(x-1)^3(x-4)^2` and zoom out. Tap the zeros: $(-3, 0)$, $(1, 0)$, $(4, 0)$. For $x < 1$ the graph is on or above the axis: statement I is true.
2. Check near $x = 4$: Type `p(3.9)` : it returns -11.6116; type `p(4.1)` : it returns -15.0176. Both are negative, so the graph touches the axis at $x = 4$ and does not cross: statement II is false.
3. Far to the left: Type `p(-10)` : it returns 12,782,924; type `p(-20)` : it returns 1,541,623,104. The values are positive and growing, so $p(x) \to \infty$: statement III is true.
4. So I and III only: option (C).

Answer: (C)

30. HARD · DESMOS

The function h is defined by $h(x) = \sqrt{x-2} + 1$. The function g is defined by $g(x) = h(x+5)$. What is the domain of g?

(A) $x \geq -3$ ✓

(B) $x \geq 2$

(C) $x \geq 7$

(D) $x \geq -5$

1. Type `h(x)=sqrt(x-2)+1` and `g(x)=h(x+5)` .
2. Tap the starting point of g: $(-3, 1)$. The graph exists only for $x \geq -3$.
3. Check: Type `g(-3)` : it returns 1; type `g(-2)` : it returns 2. (Typing $g(-4)$ gives no value.)
4. The domain is $x \geq -3$: option (A).

Answer: (A)

31. HARD · NO CALCULATOR NEEDED

A sequence is defined recursively by $a_1 = 5$ and $a_n = 2a_{n-1} - 3$ for $n \geq 2$. What is the value of a_4?

1. Type the rule as a function: `f(x)=2x-3` .
2. Start with $a_1 = 5$: Type `f(5)` : it returns 7; type `f(f(5))` : it returns 11 (these are a_2 and a_3).
3. Then Type `f(f(f(5)))` : it returns 19, which is a_4.

Answer: 19

32. HARD · DESMOS

A water-treatment filter reduces the concentration of a contaminant by 12% every hour it operates. If the initial concentration is 250 parts per million (ppm), which of the following functions models the concentration $C(t)$, in ppm, after t hours of operation?

(A) $C(t) = 250(1.12)^t$ (B) $C(t) = 250 - 12t$

(C) $C(t) = 250(0.12)^t$ **(D) $C(t) = 250(0.88)^t$ ✓**

1. After one hour a 12% decrease leaves Type `250*(1-0.12)` : it returns 220.
2. Test each option at $t = 1$: Type `250*0.12^1` : it returns 30; type `250*0.88^1` : it returns 220; type `250*1.12^1` : it returns 280; type `250-12*1` : it returns 238.
3. Only $250(0.88)^t$ gives 220. Check $t = 2$: Type `220*0.88` : it returns 193.6; type `250*0.88^2` : it returns 193.6. Option (D).

Answer: (D)

33. HARD · DESMOS

In a sequence, the first term is -4. Each term after the first is two less than triple the preceding term. Which term is the first one whose absolute value is greater than 150?

(A) 3 (B) 4

(C) 5 ✓ (D) 6

1. Type the rule: `f(x)=3x-2` .
2. Starting from $a_1 = -4$: Type `f(-4)` : it returns -14; type `f(f(-4))` : it returns -44 (a_2 and a_3).
3. Type `f(f(f(-4)))` : it returns -134; type `f(f(f(f(-4))))` : it returns -404 (a_4 and a_5).
4. $|a_4| = 134 < 150$ but $|a_5| = 404 > 150$, so the 5th term: option (C).

Answer: (C)

34. HARD · DESMOS

The exponential function $f(x) = ab^x$, where a and b are positive constants, satisfies $f(2) = 18$ and $f(5) = 486$. What is the value of $f(4)$?

1. Divide: $b^3 = \dfrac{486}{18}$. Type `(486/18)^(1/3)` : it returns 3, so $b = 3$.
2. Then $a \cdot 3^2 = 18$: Type `18/3^2` : it returns 2, so $a = 2$.
3. Check $f(5)$: Type `2*3^5` : it returns 486.
4. Finally Type `2*3^4` : it returns 162.

Answer: 162

35. HARD · DESMOS

The size of a bacterial colony, in thousands of cells, is modeled by $V(t) = 500 \cdot 2^{t/2} \cdot 4^{t/3} \cdot 16^{-t/12}$, where t is the number of hours since the start of observation. The size of the colony is multiplied by a constant factor c every 6 hours. What is the value of c?

1. Type `V(t)=500*2^(t/2)*4^(t/3)*16^(-t/12)`.

2. Compare V six hours apart: Type `V(6)/V(0)`: it returns 32.

3. Try a different start: Type `V(10)/V(4)`: it returns 32. The factor is constant, $c = 32$.

Answer: 32

PART B · SELF-ASSESSMENT — 19 PROBLEMS

36. MEDIUM · NO CALCULATOR NEEDED

What is the solution to $\sqrt{2x+7} = 5$?

1. Type `y=sqrt(2x+7)` and `y=5`.

2. Tap the intersection: $(9, 5)$, so $x = 9$.

3. Check: Type `sqrt(2*9+7)`: it returns 5.

Answer: 9

37. MEDIUM · DESMOS

The estimated population of a new coastal town t years after 2024 is modeled by

$$P(t) = 5000\,(1.15)^t\,.$$

According to the model, by approximately what factor does the population increase every 5 years?

(A) 1.15 (B) 1.75

(C) 2.01 ✓ (D) 2.99

1. Type `P(t)=5000*1.15^t`.

2. Compare five years apart: Type `P(5)/P(0)`: it returns 2.0114.

3. Try another start: Type `P(12)/P(7)`: it returns 2.0114: the same factor, so it is $1.15^5 \approx 2.01$. Option (C).

Answer: (C)

38. Medium · Desmos

A car is purchased for \$120,000. Its value depreciates according to

$$V(t) = 120000\,(0.88)^t\,,$$

where t is the number of years since purchase. To the nearest thousand dollars, what is the value of the car after 4 years?

(A) \$61,000

(B) \$72,000 ✓

(C) \$84,000

(D) \$94,000

1. Type `V(t)=120000*0.88^t`.

2. Type `V(4)`: it returns 71,963, which rounds to 72,000 dollars: option (B).

3. Trap check (simple 12% a year): Type `120000*(1-0.12*4)`: it returns 62,400, which is not an option.

Answer: (B)

39. Medium · No calculator needed

The storage capacity of a battery facility is modeled by

$$E(t) = 1450\,(1.08)^t\,,$$

where $E(t)$ is the capacity, in megawatt-hours, and t is the number of years after the facility opened, for $0 \le t \le 15$. Which is the best interpretation of 1450 in this context?

(A) The predicted capacity, in megawatt-hours, 15 years after the facility opened

(B) The estimated capacity, in megawatt-hours, when the facility opened ✓

(C) The annual percent increase in capacity

(D) The number of years it takes the capacity to double

1. Type `E(t)=1450*1.08^t`.

2. The start of the model is $t = 0$: Type `E(0)`: it returns 1,450. So 1450 is the capacity when the facility opened.

3. The other readings fail: Type `E(15)`: it returns 4,599.6452; type `ln(2)/ln(1.08)`: it returns 9.0065 (the capacity at $t = 15$ and the doubling time), neither is 1450. Option (B).

Answer: (B)

40. Medium · Desmos

A population of bacteria in a lab culture is modeled by $P(t) = 300(1.5)^t$, where t is the time in hours. What is the population after 3 hours?

(A) 675

(B) 900

(C) 1,012.5 ✓

(D) 1,350

1. Type `P(t)=300*1.5^t`.

2. Type `P(3)`: it returns 1,012.5.

3. Trap check (linear growth): Type `300*1.5*3`: it returns 1,350. The answer is option (C).

Answer: (C)

41. Medium · Desmos

The number of subscribers to an online tutoring platform grows according to $N(t) = 500(1.2)^t$, where t is measured in months since launch. Which of the following is closest to the number of subscribers after 5 months?

(A) 1,000

(B) 1,100

(C) 1,244 ✓

(D) 1,500

1. Type `N(t)=500*1.2^t`.
2. Type `N(5)` : it returns 1,244.16.
3. The closest option is 1,244 (linear trap: type `500*1.2*5` : it returns 3,000). Option (C).

Answer: (C)

42. Medium · Desmos

The amount of a medication remaining in a patient's bloodstream is modeled by $M(t) = 200(0.85)^t$, where t is the number of hours since the dose was taken and $M(t)$ is measured in milligrams. Which of the following is closest to the amount remaining after 6 hours?

(A) 65.5

(B) 75.5 ✓

(C) 85.5

(D) 95.5

1. Type `M(t)=200*0.85^t`.
2. Type `M(6)` : it returns 75.4299.
3. The closest option is 75.5 (linear trap: type `200*0.85*6` : it returns 1,020). Option (B).

Answer: (B)

43. Medium · Desmos

The population of a small town has grown at an average rate of 2.4% per year since 2010. There were approximately 60,000 residents in 2010. Which of the following functions best models the town's population P, in thousands of residents, t years after 2010?

(A) $P(t) = 60(1.024)^t$ ✓

(B) $P(t) = 60(2.4)^t$

(C) $P(t) = 1.024t + 60$

(D) $P(t) = 2.4t + 60$

1. One year of 2.4% growth on 60 thousand: Type `60*(1+2.4/100)` : it returns 61.44.
2. Test each option at $t = 1$: Type `60*1.024^1` : it returns 61.44; type `60*2.4^1` : it returns 144; type `1.024*1+60` : it returns 61.024; type `2.4*1+60` : it returns 62.4.
3. Only $60(1.024)^t$ gives 61.44. Check $t = 2$: Type `61.44*1.024` : it returns 62.9146; type `60*1.024^2` : it returns 62.9146. Option (A).

Answer: (A)

44. Medium · No calculator needed

Each term in the sequence below is 6 times the previous term. What is the 9th term in the sequence?

$$3,\ 18,\ 108,\ 648,\ \ldots$$

(A) 648×9

(B) 3^9

(C) 3×6^8 ✓

(D) 6×3^8

1. Type the rule: `a(n)=3*6^(n-1)`.
2. Check against the list: Type `a(2)`: it returns 18; type `a(4)`: it returns 648.
3. The 9th term: Type `a(9)`: it returns 5,038,848. Compare the options: Type `3*6^8`: it returns 5,038,848; type `648*9`: it returns 5,832; type `3^9`: it returns 19,683; type `6*3^8`: it returns 39,366.
4. Only 3×6^8 matches: option (C).

Answer: (C)

45. Medium · Desmos

The first term of a geometric sequence is 2 and the fourth term is 54. What is the sum of the first five terms of the sequence?

1. The fourth term is $2r^3 = 54$: Type `(54/2)^(1/3)`: it returns 3, so $r = 3$.
2. Check: Type `2*3^3`: it returns 54.
3. Sum of five terms: Type `2*(3^5-1)/(3-1)`: it returns 242.

Answer: 242

46. Hard · Desmos

The function $P(t) = \dfrac{120}{1 + 5(0.8)^t}$ models the number of active users of an app, in thousands, t weeks after launch, for $0 \le t \le 10$. Consider the following statements.
I. P is increasing on the interval $0 \le t \le 10$.
II. There is a time t in the interval at which $P(t) = 60$.
III. There is a time t in the interval at which $P(t) = 80$.
Which of the statements must be true?

(A) I only

(B) I and II only ✓

(C) II and III only

(D) I, II, and III

1. Type `y=120/(1+5*0.8^x){0<=x<=10}` (letting x stand for t). The curve rises from (0, 20) to (10, 78.0807), so statement I is true.
2. Add `y=60` and tap the intersection: (7.2126, 60). It lies inside the interval: statement II is true.
3. Add `y=80`: it never meets the graph for $0 \le t \le 10$, because the highest point is Type `120/(1+5*0.8^10)`: it returns 78.08. Statement III is false.
4. So I and II only: option (B).

Answer: (B)

47. HARD · DESMOS

The value V, in dollars, of a piece of lab equipment t years after it was purchased is modeled by

$$V(t) = 2400\,(0.64)^{t/2.5}.$$

Which of the following is the best interpretation of the model?

(A) It falls 36% every 2.5 years. ✓

(B) It falls 36% every 5 years.

(C) It falls 64% every 2.5 years.

(D) It falls 2.5% every year.

1. Type `V(t)=2400*0.64^(t/2.5)` .
2. Ratio over 2.5 years: Type `V(2.5)/V(0)` : it returns 0.64, a fall of type `(1-V(2.5)/V(0))*100` : it returns 36 percent.
3. Ratio over 5 years: Type `V(5)/V(0)` : it returns 0.4096, a fall of type `(1-V(5)/V(0))*100` : it returns 59 percent, not 36; each year: type `(1-V(1)/V(0))*100` : it returns 16.3 percent, not 2.5.
4. So it falls 36% every 2.5 years: option (A).

Answer: (A)

48. HARD · DESMOS

The number of bacteria in a colony t years after January 1, 2020 is modeled by $A(t) = 300 \cdot 4^{t/3}$. Which of the following gives the number of bacteria m months after January 1, 2023?

(A) $300 \cdot 2^{m/18}$

(B) $1200 \cdot 2^{m/36}$

(C) $900 \cdot 2^{m/18}$

(D) $1200 \cdot 2^{m/18}$ ✓

1. Type `A(t)=300*4^(t/3)` and the shifted version `B(m)=A(3+m/12)` (m months after January 1, 2023 is $t = 3 + \frac{m}{12}$).
2. At $m = 0$ and $m = 12$: Type `B(0)` : it returns 1,200; type `B(12)` : it returns 1,904.9.
3. Test each option at $m = 12$: Type `300*2^(12/18)` : it returns 476.2; type `1200*2^(12/36)` : it returns 1,511.9; type `900*2^(12/18)` : it returns 1,428.7; type `1200*2^(12/18)` : it returns 1,904.9.
4. Only $1200 \cdot 2^{m/18}$ matches: option (D).

Answer: (D)

49. Hard · Desmos

$$T(t) = 24 + 66(0.85)^t$$

The function T models the temperature, in degrees Celsius, of a cup of tea t minutes after it is placed in a room. Which of the following is the best interpretation of the number 0.85 in this context?

(A) T falls 15% per minute

(B) the gap $T - 24$ falls 15% per minute ✓

(C) $T(1) = 0.85\,T(0)$

(D) T levels off at 85% of $T(0)$

1. Type `T(t)=24+66*0.85^t`. Type `T(0)`: it returns 90; type `T(1)`: it returns 80.1.

2. The gap above 24: Type `T(0)-24`: it returns 66; type `T(1)-24`: it returns 56.1; type `0.85*(T(0)-24)`: it returns 56.1. The gap shrinks by the factor 0.85 each minute.

3. But Type `0.85*T(0)`: it returns 76.5 is not $T(1)$, so T itself does not fall by 15%. Option (B).

Answer: (B)

50. Hard · Desmos

The number of visitors to a national park increased by 25% from 2022 to 2023. From 2023 to 2024, the number of visitors decreased by 12%. If the number of visitors in 2024 was 9,020,000, what was the number of visitors in 2022?

1. Successive changes multiply: Type `1.25*0.88`: it returns 1.1, so $1.1x = 9{,}020{,}000$.

2. Solve: Type `9020000/1.1`: it returns 8,200,000.

3. Check forward: Type `8200000*1.25*0.88`: it returns 9,020,000.

Answer: 8,200,000

51. Medium · Desmos

A jacket is discounted by 30%. A 5% sales tax is then applied to the discounted price. If the final price paid is \$44.10, what was the original price of the jacket, before the discount?

(A) \$42.00

(B) \$58.80

(C) \$60.00 ✓

(D) \$63.00

1. Discount then tax multiply: Type `0.7*1.05`: it returns 0.735, so $0.735x = 44.10$.

2. Solve: Type `44.10/0.735`: it returns 60.

3. Check forward: Type `60*0.7*1.05`: it returns 44.1. Option (C).

Answer: (C)

52. Medium · Desmos

A laptop's price increased by 20% during a promotional period, then decreased by 15% after the promotion ended. If the final price is \$1,020, what was the price at the end of the promotional period, before the decrease?

(A) \$950

(B) \$1,000

(C) \$1,050

(D) \$1,200 ✓

1. The final price is 85% of the price at the end of the promotion: Type `1020/0.85` : it returns 1,200.

2. Check: Type `1200*0.85` : it returns 1,020.

3. The original price (the trap in 1,000): Type `1200/1.2` : it returns 1,000. The answer is option (D).

Answer: (D)

53. Hard · Desmos

The number of families visiting a city festival's family zone increased by 40% from Week 1 to Week 2, then decreased by 25% from Week 2 to Week 3. If Week 3 attendance was 8,400 families, how many more families attended in Week 2 than in Week 3?

1. Week 3 is 75% of Week 2: Type `8400/0.75` : it returns 11,200 families in Week 2.

2. The difference: Type `8400/0.75-8400` : it returns 2,800.

3. Check with Week 1: Type `8400/(1.4*0.75)` : it returns 8,000; type `1.4*8000` : it returns 11,200 in Week 2.

Answer: 2,800

54. Medium · No calculator needed

Which of the following represents the result of decreasing the quantity y by 12%, where $y > 0$?

(A) $0.88y$ ✓

(B) $0.12y$

(C) $y - 12$

(D) $y - 0.12$

1. Take $y = 50$. A 12% decrease: Type `50*(1-0.12)` : it returns 44.

2. Test the options at $y = 50$: Type `0.88*50` : it returns 44; type `0.12*50` : it returns 6; type `50-12` : it returns 38; type `50-0.12` : it returns 49.88.

3. Only $0.88y$ gives 44. Check $y = 200$: Type `200*(1-0.12)` : it returns 176; type `0.88*200` : it returns 176. Option (A).

Answer: (A)